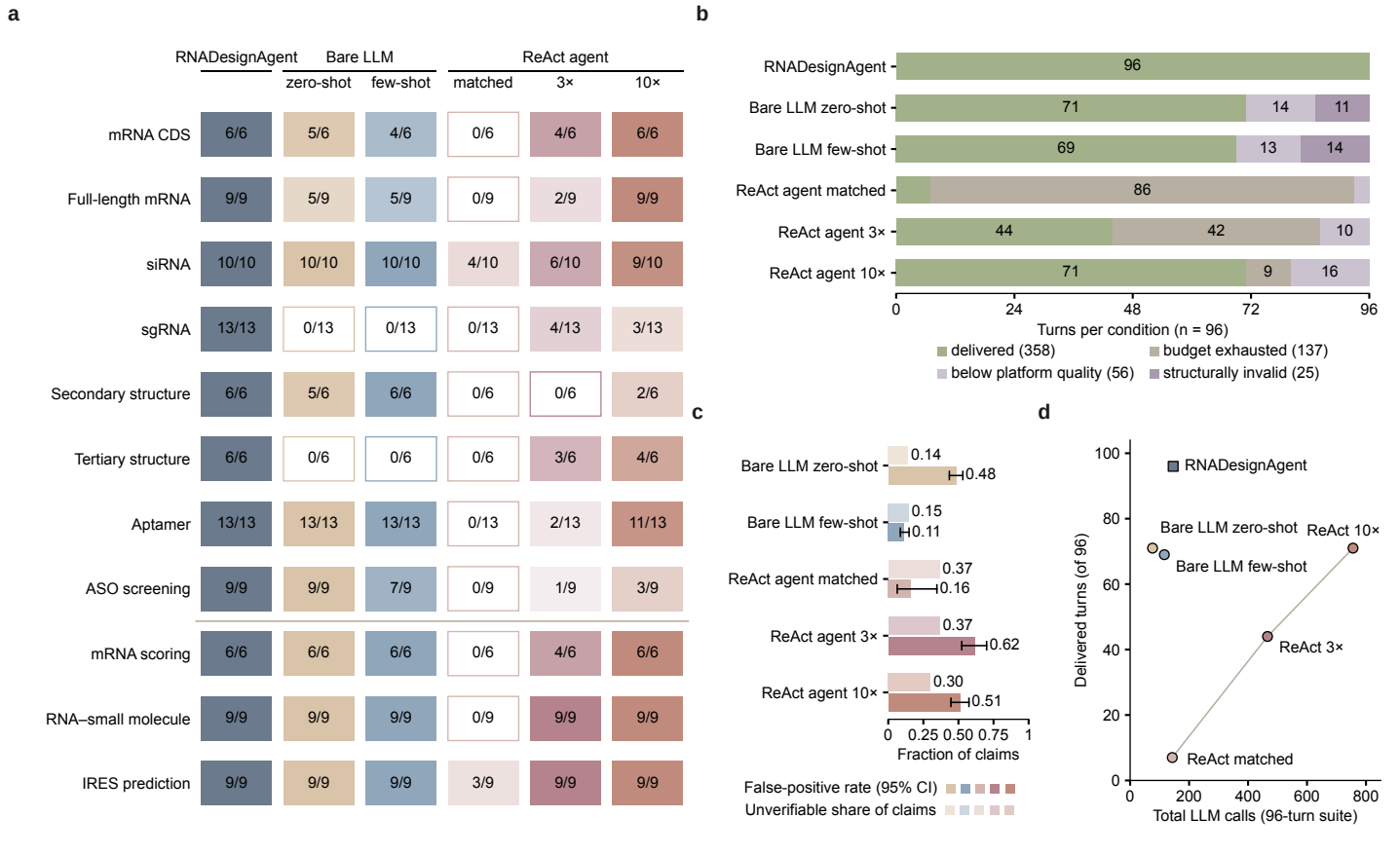

a
RNADesignAgent
Bare LLM
zero-shot
few-shot
ReAct agent
matched
3×
10×
mRNA CDS 6/6 5/6 4/6 0/6 4/6 6/6
Full-length mRNA 9/9 5/9 5/9 0/9 2/9 9/9
siRNA 10/10 10/10 10/10 4/10 6/10 9/10
sgRNA 13/13 0/13 0/13 0/13 4/13 3/13
Secondary structure 6/6 5/6 6/6 0/6 0/6 2/6
Tertiary structure 6/6 0/6 0/6 0/6 3/6 4/6
Aptamer 13/13 13/13 13/13 0/13 2/13 11/13
ASO screening 9/9 9/9 7/9 0/9 1/9 3/9
mRNA scoring 6/6 6/6 6/6 0/6 4/6 6/6
RNA–small molecule 9/9 9/9 9/9 0/9 9/9 9/9
IRES prediction 9/9 9/9 9/9 3/9 9/9 9/9
b
RNADesignAgent 96
Bare LLM zero-shot 71 14 11
Bare LLM few-shot 69 13 14
ReAct agent matched 86
ReAct agent 3× 44 42 10
ReAct agent 10× 71 9 16
0 24 48 72 96
Turns per condition (n = 96)
delivered (358)
budget exhausted (137)
below platform quality (56)
structurally invalid (25)
c
Bare LLM zero-shot 0.14 0.48
Bare LLM few-shot 0.15 0.11
ReAct agent matched 0.37 0.16
ReAct agent 3× 0.37 0.62
ReAct agent 10× 0.30 0.51
0 0.25 0.50 0.75 1
Fraction of claims
False-positive rate (95% CI)
Unverifiable share of claims
d
RNADesignAgent
Bare LLM zero-shot
Bare LLM few-shot
ReAct 10×
ReAct 3×
ReAct matched
Delivered turns (of 96)
0 20 40 60 80 100
0 200 400 600 800
Total LLM calls (96-turn suite)